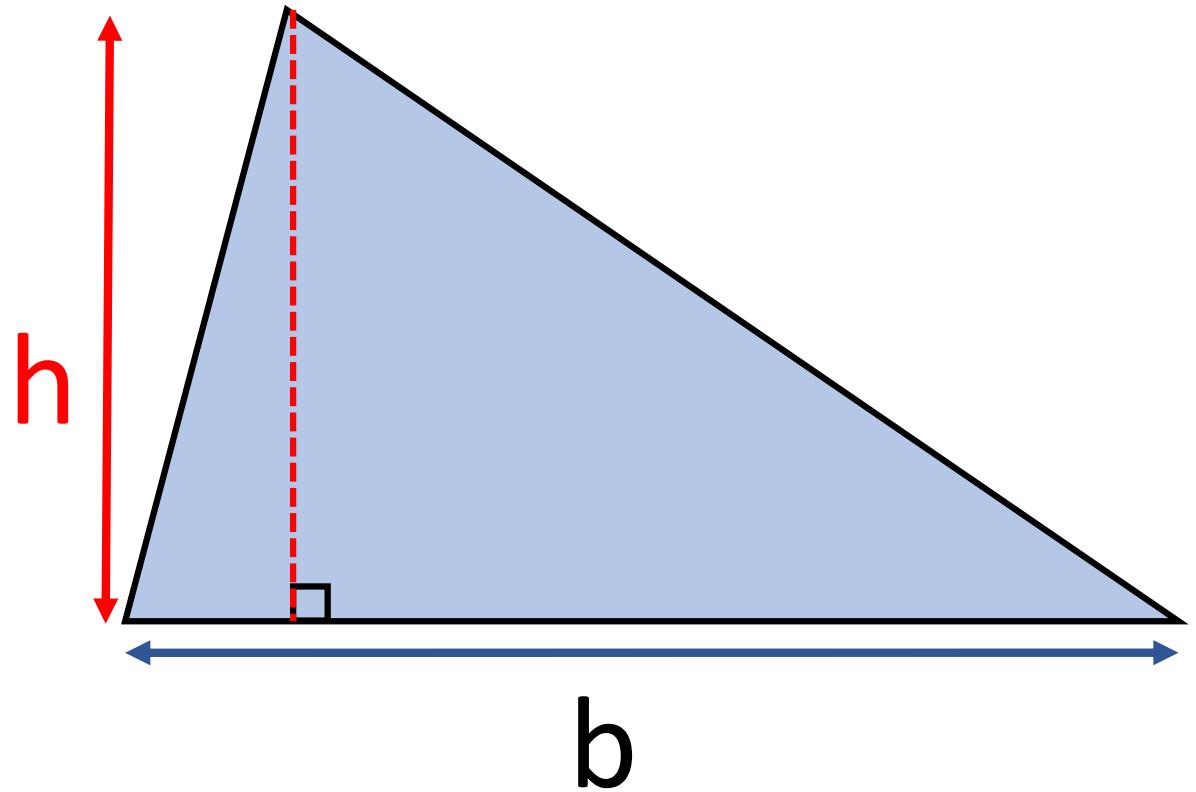
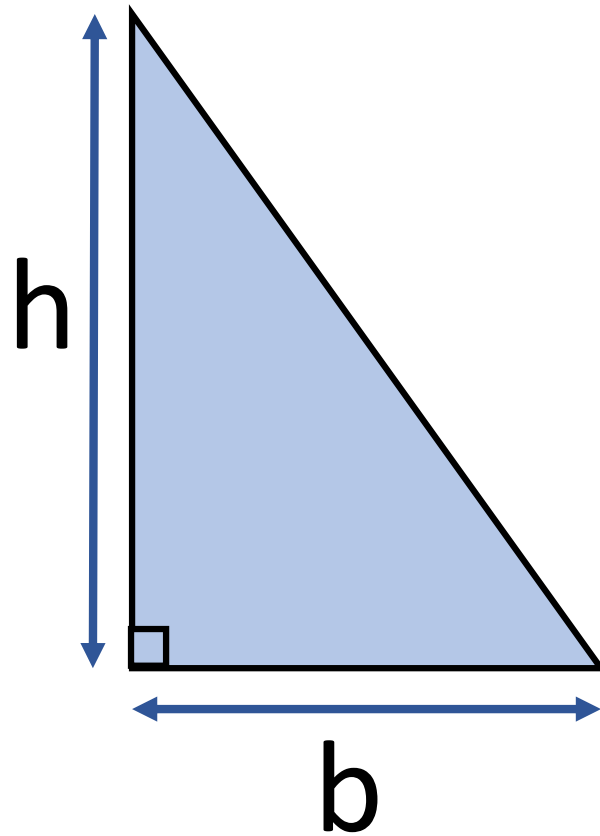
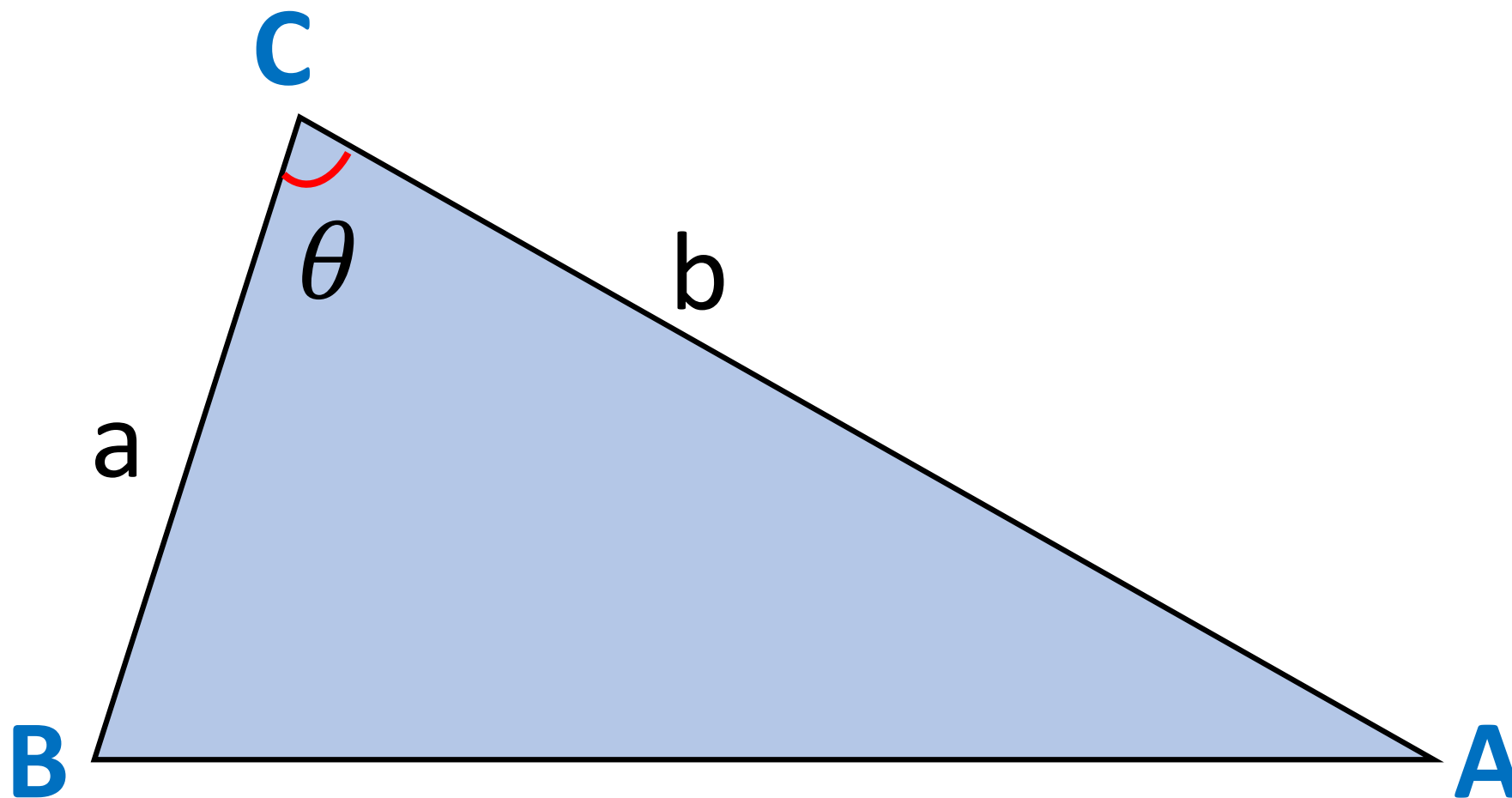


Area of a Triangle

Area of a triangle

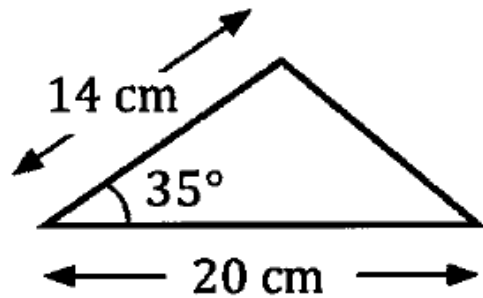


$$\text{area of a } \Delta = \frac{1}{2} b h$$



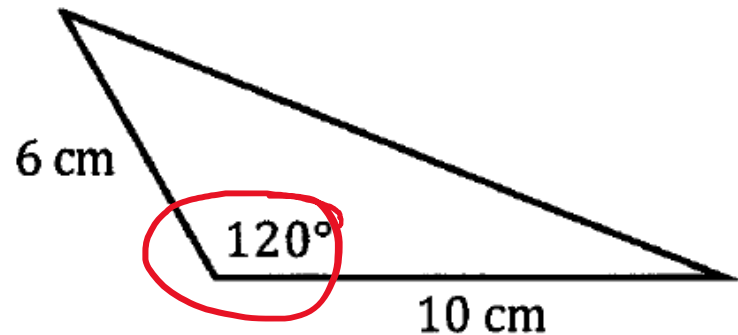
$$\text{area of a } \Delta = \frac{1}{2} a b \sin \theta$$

Examples



$$\begin{aligned} A &= \frac{1}{2}(20)(14)\sin 35^\circ \\ &= 140 \sin 35^\circ \\ &= 80.30 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \sin \theta &= \sin(180^\circ - \theta) \\ \sin 60^\circ &= \sin 120^\circ \end{aligned}$$



$$\begin{aligned} \text{Area} &= \frac{1}{2}(6)(10)\sin 120^\circ \\ &= 30 \sin 120^\circ \\ &= \sim \sim \sim \sim \end{aligned}$$

Find the angle C for the ΔABC given that $a = 4$ cm, $b = 3$ cm and the area of the $\Delta ABC = 3$ cm².

$$A = 3 \text{ cm}^2$$

$$3 = \frac{1}{2}(4)(3) \sin \theta$$

$$3 = 6 \sin \theta$$

$$\sin \theta = \frac{1}{2}$$

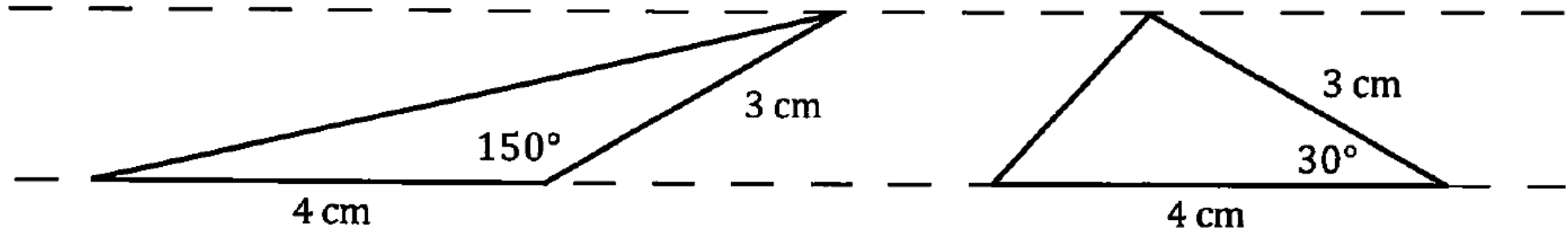
$$\theta = \sin^{-1}\left(\frac{1}{2}\right)$$

$$\theta = 30^\circ \text{ or } 150^\circ$$

2 possible
solutions

because

$$\sin \theta = \sin(180^\circ - \theta)$$

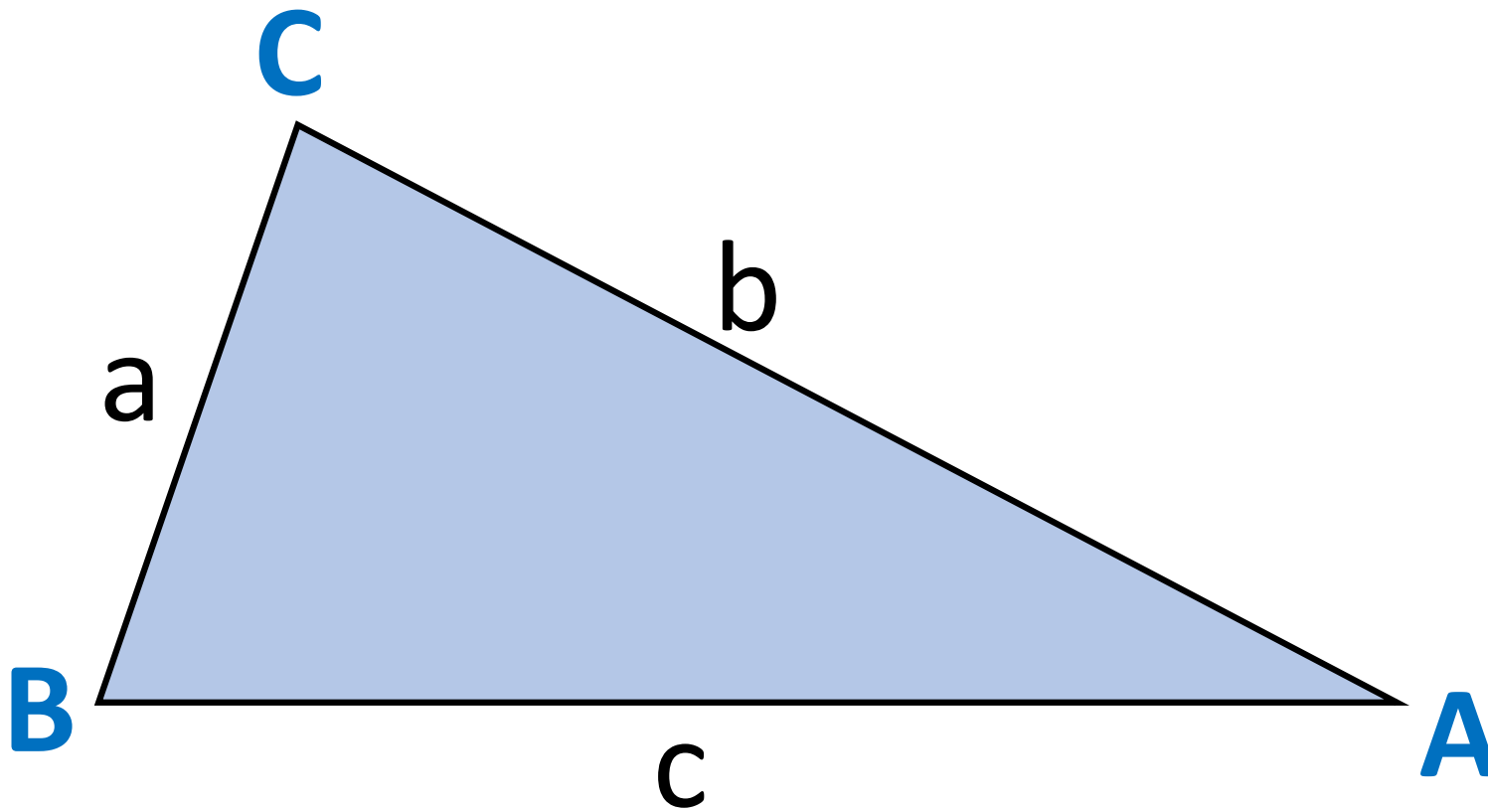


ΔABC is such that $a = 16$ cm, $b = 12$ cm. The area of the $\Delta ABC = 78$ cm². Find the angle A .

$$78 = \frac{1}{2} (16)(12) \sin A$$

$$A = 54.34^\circ \text{ or } 125.66^\circ$$

Generally you should be told if it is an acute or obtuse Δ .

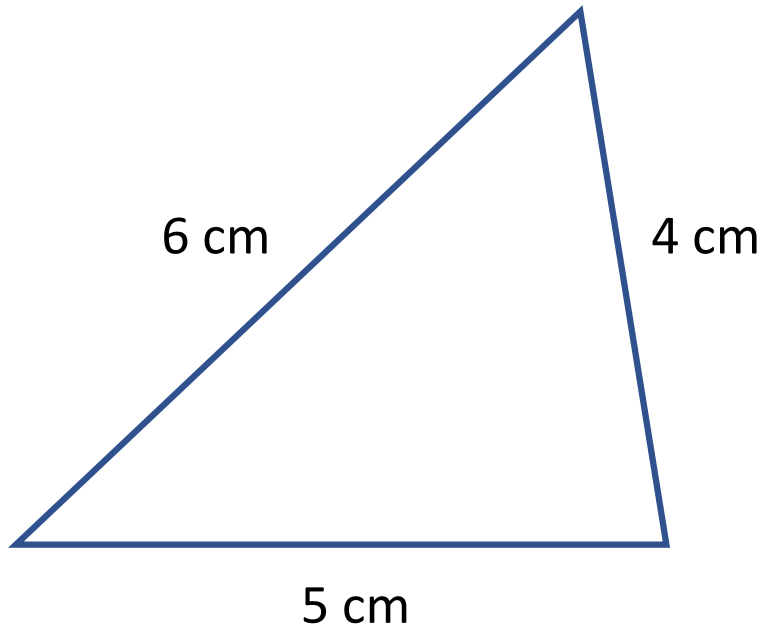


Heron's formula:

$$\text{area of a } \Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

Where

$$s = \frac{a+b+c}{2}$$



$$s = \frac{6 + 4 + 5}{2} = 7.5$$

$$A = \sqrt{7.5(7.5 - 6)(7.5 - 4)(7.5 - 5)}$$
$$= 9.92 \text{ cm}^2 \checkmark$$

Find the area of a $\triangle ABC$, two sides of which are 8 cm and 11 cm and the perimeter is 32 cm.

$$\text{third side} = 32 - (11 + 8) = 13$$

$$s = \frac{32}{2} = 16$$

$$A = \sqrt{16(16-13)(16-8)(16-11)}$$
$$= 43.82 \text{ cm}^2$$